

Science

Year 2

Student Manual



Science

Year 2 · Student Book

How do you know?

INSIDE THIS BOOK

- Twenty-five topics across six units
- Over 350 things to do, say and draw
- A modelled enquiry in every single topic
- Six projects, and a garden guide to finish
- QR codes that open real science beyond the page

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Ages 6 to 7 · Lower Primary

www.primeschool.pt

Come into the shed



The potting shed, early, before anyone has come in.

At the top of the garden there is an old wooden shed. Inside there is a long bench and a stack of pots. Two watering cans stand by the window, and the sill is full of seedlings.

Six friends work in this shed. They grow things and mend things. They build things and test things. When something surprises them, they write it down.

That is what a scientist does. You are going to do it too.

This book is yours to write in. Draw in it. Fill in the boxes. Cross things out if you change your mind. A scientist does that too, every time she finds something new.



Prime School

for teachers and families

primeschool.pt

The six in the shed



Bram, Rook, Moss, Nib, Fen and Tick.

Each one of the six is best at one kind of science. When you meet a new unit, you will meet the friend who knows most about it.



Bram

the badger. He sows the seeds.



Rook

the rook. He pushes and pulls.



Moss

the hedgehog. She knows every material.



Nib

the wood mouse. He waters and measures.



Fen

the toad. He carries the light.



Tick

the wren. She makes the bell ring.

Bram and Nib are the two on the front of this book.

What is in this book



Six pots on the windowsill, one for each unit.



What is in this book, continued

At the back: our own garden guide, the science skills you have learned, and a word list of everything you have been taught.

What the boxes mean



Everything you need is already on the bench.

On the bench	A question to start you off. Answer it out loud.
What Bram noticed	Something one of the six saw, and could not explain yet.
By the end you will	What you will be able to do when the unit is finished.
Wake up your thinking	A quick game before the work starts.
Do you remember?	Something you already know, needed again here.
Let us find out	The new science. Read this part slowly.
Watch me find out	One of the six does an experiment and shows you the answer.
Your turn	Numbered work for you to do. Write or draw in the space.
How do you know?	Say why you think so. This is the most important box.

And these ones too

The material you need	Fetch these before you start.
Find out together	Work with a partner or in a group.
Take care	Stop and read this. An adult must be with you.
Nib's notebook	Something surprising that is true.
Look a little closer	Extra science, if you want to go further.
Potting shed challenge	A hard one. Have a go even if you are not sure.
Along the way	One step of the big project you build all year.
Words we used	The new words from this unit, with pictures.
Now I can	Tick off what you can do at the end of a unit.
Check what you can do	Five jobs that show what you learnt in the unit.
How did you do?	Say how sure you feel. There is no wrong answer here.



What Year 2 science covers

for teachers

primeschool.pt, Prime School curriculum

▲ For teachers

This book carries a printed web address beside every QR code, so a class with no devices loses nothing: the code is a shortcut, never the only route. Codes are addressed to adults, and each panel says so.

Pupils should be shown how to read the feature labels before Unit 1. The labels are consistent across all six units, so a pupil who learns them once can navigate the book independently for the rest of the year.

What a scientist needs



The kit you will use again and again this year.

Everything you need is on this page. Most of it is already in the classroom, and the rest you can find outside.



hand lens

makes small things look bigger



jar

for holding what you find



ruler

for measuring how long



dish

for water, soil or seeds



seed tray

for growing things in



watering can

for giving plants the same drink



magnet

for finding out what it pulls



torch

for making light in the dark

▲ The material you need

a hand lens, a jar, a ruler, a seed tray and a pencil

! Take care

Wash your hands after digging or holding animals. Put every living thing back exactly where you found it.

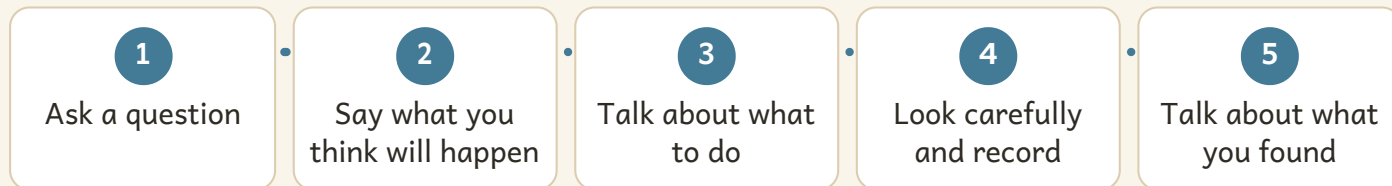
How we find things out



Nib grew one bean four times, and measured it each week.

Scientists do not only look. First they ask a question. Then they find out. Then they say what they found.

The five steps we use every time



You will use these five steps in every unit. They are the steps every scientist uses.

Do not worry if what happens is not what you thought. That happens to scientists all the time. It is how they learn something new.

GETTING SET UP

Looking closely, and being fair



An ivy leaf, and the same leaf through Bram's hand lens.

GETTING SET UP · CONTINUED

A hand lens makes small things look bigger. Hold it close to the thing, then move it slowly until you can see clearly.

A test is **fair** when you change one thing only. If you give one plant more water, give it nothing else extra.

Find out together

Look at your own thumb through the hand lens. Then look at a leaf. Then look at a grain of soil. Which one surprised you most?

▲ The material you need

a hand lens, an ivy leaf and a few grains of soil



Safe practical science in a primary classroom

for teachers

ase.org.uk, primary science

UNIT 1

Environments and habitats

Bram has lifted a flat stone at the edge of the bed. Something under it is still moving.



On the bench

Look out of the window. Can you see a place where something lives? How do you know something lives there?

○ What Bram noticed

I lifted the flat stone by the bed, and there were three woodlice under it. Why were they all under that one stone?

Bram is right to wonder. There are stones all over the garden. Something about that one suited a woodlouse.

● By the end you will

- say what a habitat is
- name plants that suit shade, dry walls and water
- match an animal to the habitat that suits it
- sort rocks by how hard they are
- say one way to care for your own environment



Four habitats in one small garden.



Four places to live in one garden

1.1 Habitats

WAKE UP
YOUR
THINKING

Point to somewhere in this room that would suit a spider.
Now point to somewhere that would not suit it at all.

▲ Let us find out

A **habitat** is the place where an animal or a plant lives. A good habitat gives it everything it needs.

Every living thing needs food and water. It needs air. It needs a safe place to rest.

The log pile is a habitat. It is dark and damp under the bark, and a woodlouse likes that. The wall is a habitat too, but it is dry and sunny. A woodlouse would soon dry up there.

So a habitat is not just any place. It is the **right** place.

★ Watch me find out 1

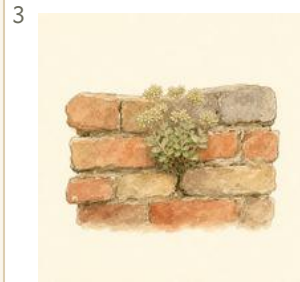
Why do the woodlice choose that stone?

- 1 Bram lifted the flat stone. It was dark and damp underneath.
- 2 He lifted a stone from the sunny path. It was dry underneath, and there were no woodlice.
- 3 A woodlouse has a soft body that dries out fast.

We found out: the woodlice chose the damp stone, because damp shade is the habitat that suits them

How do you know?

Nib says the log pile and the brick wall are the same. Both are made of hard stuff, he says. Is he right? How do you know?

Your turn

Four habitats in the garden: a log pile, a pond, a wall and a bed.

- 1 Write the name under each habitat above.
- 2 Which of the four is the wettest?
- 3 Which of the four is the driest?
- 4 Name one thing every habitat must give an animal.
- 5 Draw the habitat you would choose if you were a woodlouse.
- 6 Say why you chose it.
- 7 Name a habitat you can see from your classroom window.

★ Nib's notebook

A single log pile can be home to more than a thousand creatures at once. Most of them are smaller than a grain of rice.

Your turn, and finding out together

A habitat gives an animal what it needs.	TRUE / NOT TRUE
Every animal likes the same habitat.	TRUE / NOT TRUE
A pond is a habitat.	TRUE / NOT TRUE
A woodlouse would be happy on a hot dry path.	TRUE / NOT TRUE

- 8 Tick TRUE or NOT TRUE for each sentence above.
- 9 Sort these into wet or dry: pond, wall, log pile, path.
- 10 Draw one animal you have really found outside.
- 11 Write where you found it.
- 12 Bram says a fish could live in the log pile. Is he right? Say why.
- 13 Name two habitats in your own garden or street.
- 14 Which habitat in this school would suit a snail best?

▲ **The material you need**

a hand lens, and a flat stone from outside with damp soil under it

Find out together

Turn over one flat stone outside. Count how many living things you find under it. Look with the hand lens. Then put the stone back exactly as it was.

! **Take care**

Put every animal back where you found it, and wash your hands afterwards.



Three plants, three very different places

1.2 Plants in different habitats

Do you remember?

You know that a habitat gives a living thing what it needs. What do you think a plant needs?

▲ Let us find out

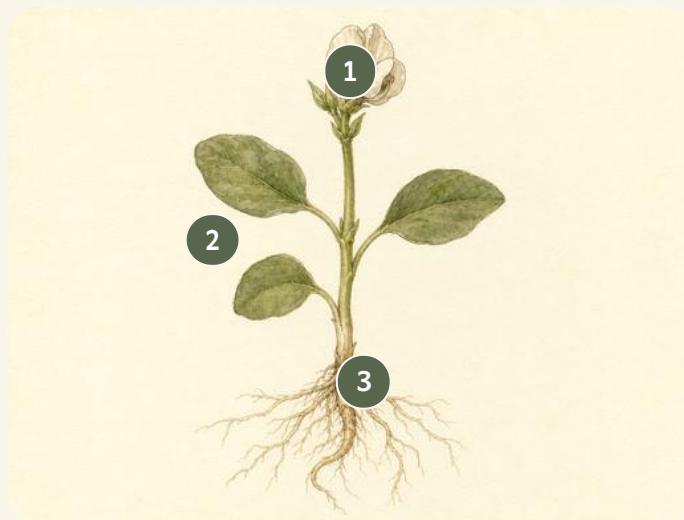
Plants live in different habitats, and each one suits where it grows.

A **fern** grows in deep shade. Its wide flat leaves catch the little light that reaches it.

Stonecrop grows in a dry crack in the wall. Its leaves are thick and fat, because they store water for the dry days.

A **reed** stands in the shallow water at the edge of the pond. Its stem is tall and hollow, so it can stand up in mud.

Move the fern to the wall and it would dry out. Each plant belongs where it is.



- | | |
|----------|----------|
| 1 flower | 4 2 leaf |
| 3 stem | 4 root |

This bean plant has 4 parts labelled. Find each one on a real plant outside.

Your turn

★ Watch me find out 2

Why are the stonecrop's leaves so fat?

- 1 Nib felt a stonecrop leaf. It was thick and full.
- 2 He felt a fern leaf. It was thin and flat.
- 3 The stonecrop grows in a dry wall. The fern grows in damp shade.

We found out: the stonecrop has fat leaves because it stores water for the dry days

- 1 Which plant grows in deep shade: the fern or the stonecrop?
- 2 Which plant stands in shallow water?
- 3 Why does the fern have wide flat leaves?
- 4 Draw a plant. Label its root and its leaf.
- 5 Which part of the plant holds it in the soil?
- 6 Which part makes the seeds?
- 7 Name a plant part you have eaten.

How do you know?

Moss moves a fern into the dry sunny wall. What do you think will happen to it? How do you know?

Your turn, and looking closely

1



2



3



Match each plant to its habitat: shade, dry wall, shallow water.

8 Write the name under each plant above.

9 Write shade, wall or water under each one.

10 Draw the leaf of a plant from outside. Draw its veins too.

11 Count the leaves on your plant.

12 Is your plant in a damp place or a dry one?

13 Find two plants growing in the same crack. Draw them both.

14 Which of your two plants is taller?

WHAT I THINK WILL HAPPEN

Write the number you think

WHAT REALLY HAPPENED

Draw or write what really happened

How many little roots will come off the big one?

▲ The material you need

a small weed pulled up gently with its root on, a dish of water and a hand lens

Find out together

Pull up one small weed with its root. Wash the soil off the root in the dish. Look at the root with the hand lens. How many little roots come off the big one?

15 Count the little roots. Write the real number next to your guess.

16 Compare the two numbers. Was it what you thought?

Look a little closer

Some plants store their food in a fat root, so we can eat it. Name two roots you have eaten.



1.3 Animals in different habitats

WAKE UP
YOUR
THINKING

Think of an animal with wings. Now think of one with no legs at all. Which one can reach more places?

▲ Let us find out

Animals live in the habitat that suits their bodies.

A **snail** keeps to the damp shade. It has a soft wet body, and the sun would dry it out.

A **frog** stays close to the pond. It needs water to lay its eggs in.

A **blackbird** can go anywhere in the garden, because it has wings and feathers to keep it warm.

When you look for an animal, think first about the place it would choose.

Bram's own notes

living thing	where you would look for it
snail	damp shade
frog	at the pond
blackbird	anywhere in the garden
woodlouse	under a stone



Look up any animal you found
for teachers and families
[wildlifetrusts.org](https://www.wildlifetrusts.org), wildlife explorer

Your turn

★ Watch me find out 3

Where should Bram look for a frog?

- 1 A frog has damp skin, so it needs a wet place.
- 2 A frog lays its eggs in water.
- 3 Bram looked at the edge of the pond, in the wet mud.

We found out: the pond edge suits a frog, so that is where Bram found one

- 1 Where would you look for a snail? Say why.
- 2 Which animal on this page has wings?
- 3 Why can a blackbird live in more places than a snail?
- 4 Draw a woodlouse. Draw the stone it lives under too.
- 5 Name one animal that needs water to lay its eggs.
- 6 Name one animal you would find in a log pile.
- 7 Which animal would you never find in the pond?

How do you know?

Tick says a woodlouse could live at the top of the wall, because a woodlouse can walk. Is she right? How do you know?

★ Nib's notebook

A garden snail can pull its whole body inside its shell. Then it closes the door with a plug of dried slime.

Your turn, and counting for real

What Nib found under one log in ten minutes

What we found	Tally	How many
snail		4
woodlouse		7
worm		3
beetle		2
altogether		16

- 8 Which animal did Nib find most of?
- 9 Which did he find fewest of?
- 10 How many animals did he find altogether?
- 11 Count the animals you can see in your own playground. Make a tally.
- 12 Draw the animal you found most of.
- 13 Say one reason your count is different from Nib's.
- 14 Would you find the same animals in winter? Say what you think.

▲ The material you need

a hand lens, a notebook and a pencil

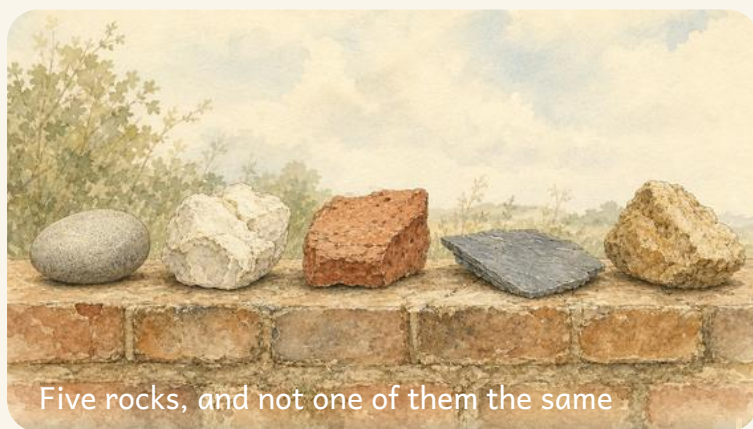
Find out together

Choose one square of ground the size of this book. Watch it for five minutes and tally every living thing that moves across it.

**Count birds for real science**

for teachers and families

rspb.org.uk, big garden birdwatch



Five rocks, and not one of them the same

1.4 Rocks and the environment

Do you remember?

You have sorted plants and animals. Now you will sort something that was never alive. What is a rock made of?

▲ Let us find out

Rocks are part of the environment too, and they are not all the same.

Some rocks are **hard**, like slate. You cannot scratch slate with your nail.

Some are **soft**, like chalk. Chalk crumbles into dust in your hand.

Rocks matter to living things. Stonecrop grows in the wall because the brick holds a little soil. A snail hides under a flat stone because it stays cool and damp there.

Moss tested four rocks with her own claw

rock	hard or soft?	what it feels like
slate	hard	smooth and flat
chalk	soft	dusty
brick	hard	rough
sandstone	soft	grainy



Where rocks come from

for teachers

bgs.ac.uk, discovering geology

Your turn

★ **Watch me find out 4**

Which rock is the softest?

- 1 Moss scratched each rock with the same iron nail.
- 2 The nail left no mark at all on the slate.
- 3 The nail went straight into the chalk and left white dust.

We found out: the chalk is the softest, because the same nail marked it most

- 1 Which rock would you use to build a wall? Say why.
- 2 Which rock crumbles in your hand?
- 3 Order these from softest to hardest: slate, chalk, brick.
- 4 Draw a rock you have found. Draw its bumps and its colour.
- 5 Is your rock rough or smooth?
- 6 Name one animal that hides under a stone.
- 7 Why is a flat stone a good place for a snail to hide?

How do you know?

Fen says chalk must be strong because a whole cliff is made of it.
What do you think? How do you know?

Your turn, and a fair test

WHAT I THINK WILL HAPPEN

Draw or write which rock you think, before you start

WHAT REALLY HAPPENED

Draw or write what really happened

Which rock will make the most dust when you rub it?

- 8** Write down which rock you think will make most dust.
- 9** Rub each rock ten times on the same paper. Use the same rubs.
- 10** Which rock made the most dust?
- 11** Was it the one you thought? Compare what really happened with your guess.
- 12** Why did you rub each rock the same number of times?
- 13** Draw the dust each rock made.
- 14** Which rock made no dust at all?

▲ The material you need

a piece of chalk, a piece of slate, a piece of brick and a sheet of dark paper

Find out together

Rub each rock the same number of times on the dark paper. Look at the dust with the hand lens. Which rock is losing the most of itself?

✦ Potting shed challenge

Rain rubs at rock for thousands of years. Which of your rocks do you think the rain would wear away first?



Two friends putting one small place right

1.5 How can we care for our environment?

WAKE UP
YOUR
THINKING

Look around you. Can you see one thing that should not be there?

▲ Let us find out

People change the environment. Some changes help, and some do harm.

Dropping litter does harm. A hedgehog can get its head stuck in a tin, and a bird can swallow plastic.

Planting a flower helps, because a bee can feed from it. Leaving a log pile alone helps, because it is somebody's home.

You cannot mend everything. But you can look after the small piece of the world you stand in.

Helps:

planting a flower, leaving a log pile,
filling a bird bath, picking up litter

Harms:

dropping litter, pulling up every
weed, frightening the birds, tipping
paint on the soil

**Look up any animal you found**

for teachers and families

wildlifetrusts.org, wildlife explorer

Your turn**★ Watch me find out 5****Does the litter really matter?**

- 1 Bram left a tin in the grass for one week.
- 2 When he came back, a beetle and two woodlice were trapped inside it.
- 3 He took the tin away and put it in the bin.

We found out: litter does matter, because animals get trapped in it

- 1 Name one thing that helps the garden.
- 2 Name one thing that harms it.
- 3 Draw a bin. Draw one thing that belongs in it.
- 4 Why should you leave a log pile alone?
- 5 Say one way your class could help the birds.
- 6 Say one way you could help at home.
- 7 Who looks after the ground outside your school?

How do you know?

Rook says pulling up every weed makes the garden better. What do you think? How do you know?

✦ Nib's notebook

One dropped plastic bag can still be in the soil fifty years later. The child who dropped it will be grown up by then.

Your turn, and doing something about it

Our class promise

- 8 Write one promise your class can really keep.
- 9 Draw the place in your school you want to look after.
- 10 Count the pieces of litter you can see there.
- 11 Pick them up with a grown-up. Count them again.
- 12 Which number is smaller?
- 13 Draw one animal that will be safer now.
- 14 Say what you will do next week.

▲ **The material you need**

gloves for everybody, one bag for litter and one bag for things that can be recycled

Find out together

Choose one small place in your school grounds. Tidy it together. Then look after it for the rest of the year.

! **Take care**

Wear gloves. Never pick up glass, or anything sharp, or anything you do not know. Tell an adult instead.

UNIT 1 • LOOKING BACK

Words we used

■ WORDS WE USED

**habitat**

the place where a plant or animal lives

**environment**

everything around a living thing

**shade**

a place the sun does not reach

**store**

to keep something for later

**shallow**

not deep

**hard rock**

you cannot scratch it easily

**soft rock**

it crumbles or scratches easily

**litter**

rubbish dropped where it does not belong

■ Check what you can do

- 1 Point to a habitat out of the window and say what lives there.
- 2 Name a plant that suits deep shade, and say how its leaves help it.
- 3 Say why a snail would not last a day on a hot dry path.
- 4 Scratch two rocks with the same nail and say which is softer.
- 5 Say one thing that harms your environment and one that helps it.

UNIT 1 • LOOKING BACK

Now I can

- ☐ say what a habitat is, and what it must give an animal
- ☐ name a plant that suits shade, a dry wall and shallow water
- ☐ match an animal to the habitat that suits its body
- ☐ sort rocks into hard and soft, and prove it with a fair test
- ☐ say how I can care for the small piece of world I stand in

How did you do?

	confident	almost there	needs more work
I can say what a habitat is	☐	☐	☐
I can match a plant to its habitat	☐	☐	☐
I can match an animal to its habitat	☐	☐	☐
I can sort rocks by how hard they are	☐	☐	☐
I can name a way to care for my environment	☐	☐	☐



Bram's map of every habitat in one small garden.

▲ For teachers

Pupils at this age readily accept that a habitat is a place, and less readily that it is the RIGHT place. The woodlouse test in Topic 1.1 is the argument that carries it: same garden, two stones, different results. Repeat it outdoors if a pupil is unconvinced.

Topic 1.4's dust test is the book's first fair test. The point is not which rock wins, it is that the number of rubs was the same for each. Ask pupils what would happen if one rock had been rubbed twice as much.

Forces and movement

Rook has left the barrow at the foot of the path. It will not stay still.



On the bench

Push this book a little way across the table. Now pull it back. You have just used two forces.

○ What Rook noticed

I put six pots in the barrow. On the flat path it rolled slowly. On the slope it ran away from me.

Rook is right to wonder. The barrow did not change. Something about the path did.

● By the end you will

- say what a force is, and point to a push and a pull
- squash, bend, twist and stretch to change a shape
- make a thing go faster or slower with a force
- change the way a moving thing is going



Every job on the path needs a push or a pull.



Every job on the path is a push or a pull

2.1 Forces around us

WAKE UP
YOUR
THINKING

Push down on the table with both hands. Now pull the edge of your chair. You have just made two forces.

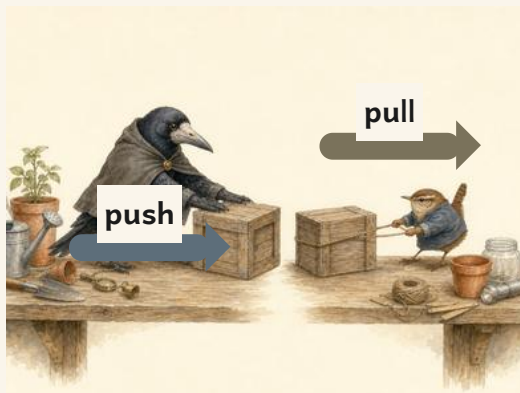
▲ Let us find out

A **force** is a push or a pull. You cannot see a force. You can see what it does.

Rook pushes the **barrow** up the path. His push makes it start to move. He pulls the handles back to stop it.

A force can make a still thing start to move. A force can make a moving thing stop.

Look for a push and a pull all round the shed. The door needs a pull. The trowel needs a push.



More things to push and pull

for teachers

rigb.org, learning activities

Your turn

★ Watch me find out 1

Does the barrow need a push or a pull?

- 1 Rook stood the loaded barrow at the foot of the path.
- 2 He pushed the handles. The barrow rolled up the path.
- 3 He pulled the handles back. The barrow stopped.

We found out: a push started the barrow, and a pull stopped it

- 1 Name one thing in this room you must pull.
- 2 Name one thing in this room you must push.
- 3 Draw the barrow. Draw an arrow to show Rook's push.
- 4 Is opening a door a push or a pull? Say which.
- 5 A ball lies still on the grass. What makes it start to roll?
- 6 A ball is rolling. What makes it stop?
- 7 Write down one force you have used today.

How do you know?

Bram says a force is always a push. What do you think? How do you know?

✦ Nib's notebook

A rook can carry a twig as long as your ruler in its bill. It flies with the twig up to the top of a tall tree.

Your turn, and finding out together

1



2



3



4



Four things in the shed. Every one needs a push or a pull.

8 Write the name under each thing above.

9 Write push or pull beside each name.

10 Draw a straight arrow to show which way each force goes.

11 Find two things outside that you must pull. Draw them both.

12 Find one thing you have to push with two hands.

13 Say what a still ball does when nobody pushes it.

14 Which needs a bigger push: the empty barrow or the full one?

▲ The material you need

a small barrow or a toy cart, three pots, a length of twine and a wooden block

Find out together

Load the barrow with three pots. Push it along a flat path. Then tie the twine on and pull it back. Which took more work, the push or the pull?

! Take care

Keep your feet away from the wheel. Never push the barrow at somebody.



2.2 Changing shape

Do you remember?

A force can start a thing moving. Can a force change a thing without moving it along?

▲ Let us find out

A force can change the shape of a thing. It does not have to move it along.

Rook **squashes** the ball of clay flat. He **bends** a green twig into a ring. He **twists** the twine round the peg. He **stretches** the rubber ring long and thin.

Squash, bend, twist and stretch are all pushes or pulls.

Some things spring back to the old shape. The rubber ring goes short again. Some things keep the new shape. The flat clay stays flat.

Four ways a force changes a shape

squash	bend	twist	stretch		
--------	------	-------	---------	--	--

Fill the two empty boxes with a shape you have changed yourself.

Your turn**★ Watch me find out 2****Which things go back to their old shape?**

- 1 Rook squashed the ball of clay flat. It stayed flat.
- 2 He stretched the rubber ring long. It went short again.
- 3 He bent a green twig into a ring. It sprang back a little way.

We found out: the clay kept its new shape. The rubber ring went back to its old one

- 1 Draw a ball of clay. Now draw it after a squash.
- 2 Which word means to pull something longer: squash or stretch?
- 3 Name one thing in the shed you could bend.
- 4 Name one thing you should never bend.
- 5 Twist a piece of twine. Draw what it looks like now.
- 6 Write squash, bend, twist or stretch beside each of these: clay, twig, twine, rubber.
- 7 Which of the four ways was hardest to do? Say why.

How do you know?

Nib says the green twig will spring all the way back. What do you think? How do you know?

Your turn, and finding out together

The clay, squashed and unsquashed, then twine and a spring.

- 8 Write the name of each thing above.
- 9 Beside each one write squash, bend, twist or stretch.
- 10 Press a ball of dough flat. Draw it before and after.
- 11 Roll the dough back into a ball. Can you do it?
- 12 Which bends better: a green twig or a dry one?
- 13 Find one bendy thing and one stiff thing in your bag.
- 14 Say what your hands did to change each shape.

▲ The material you need

a lump of clay or dough, a green twig, a length of twine and a rubber ring

Find out together

Take a lump of dough each. Squash it, bend it, twist it and stretch it. Draw the shape you like best. Then roll it back into a ball.

✦ Potting shed challenge

A bent nail stays bent. A rubber ring does not. Find three more things for each group.



The ramp, the barrow and one very fast wheel

2.3 Changing speed

WAKE UP
YOUR
THINKING

Roll a pencil gently across the table. Now roll it hard.
Which roll went further?

▲ Let us find out

A bigger push makes a thing go faster. A small push makes it go slowly.

Rook pushes the barrow hard, and it runs up the path. He pushes it gently, and it hardly moves.

A **slope** changes the **speed** too. A ball rolls faster down a slope. It slows down going up.

Friction is the rub between two things. A rough path rubs the wheel and slows it down. A smooth path lets the wheel run on.

Rook's own notes from the ramp

how Rook pushed the ball	how far the ball went
a small push	half way along the bench
a big push	right to the end
a big push, on rough cloth	it stopped early

Your turn**★ Watch me find out 3****Does a bigger push make the ball go faster?**

- 1 Rook let the wooden ball roll down the ramp with no push.
- 2 It rolled to the middle of the bench and stopped.
- 3 Then he gave it a big push. It shot off the end.

We found out: a bigger push made the ball go faster and further

- 1 Give a ball a small push. Mark where it stops.
- 2 Now give the same ball a big push. Mark where it stops.
- 3 Which push sent the ball further?
- 4 Draw your two marks on the same line.
- 5 Say what makes a rolling ball slow down.
- 6 Which is faster: a ball on a slope or a ball on the flat?
- 7 Name one place in school where the ground is rough.

How do you know?

Bram says the ball stops because it gets tired. What do you think?
How do you know?

★ Nib's notebook

A wheel rolls best on something smooth. A wet path has less rub, so a loaded barrow can slide away from you.

Your turn, and a fair test

WHAT I THINK WILL HAPPEN

Draw or write which floor you think, before you start

WHAT REALLY HAPPENED

Draw or write what really happened

Which floor will let the ball roll furthest?

- 8** Write which floor you think will let the ball roll furthest.
- 9** Roll the same ball on wood, on cloth and on stone.
- 10** Push the ball the same way every time.
- 11** Measure how far the ball went each time.
- 12** Which floor let the ball roll furthest?
- 13** Was it the one you thought? Compare what really happened with your guess.
- 14** Say why you used the same ball every time.

▲ The material you need

a wooden ball, a strip of cloth, a smooth board, a ruler and a piece of chalk

Find out together

Lay the cloth over one half of the bench. Roll the ball from the same place on to each side. Mark where the ball stops each time.

✦ Potting shed challenge

Lift the end of the ramp a little higher. Does the ball go faster? Try it three times.



2.4 Changing direction

Do you remember?

You know that a force can start, stop and speed up a moving thing. Can a force turn it?

▲ Let us find out

A force can change the way a moving thing is going. We call that its **direction**.

Rook rolls the ball straight down the bench. Bram taps it from the side. The ball turns and rolls another way.

The tap was a push. A push from the side turns a moving thing.

You do this every day. You turn the barrow round a corner with your hands. You bat a ball back with your hand.

A wall changes direction too. A ball thrown at a wall comes back at you.

A push from behind:

the ball goes faster along the same line.

A push from the side:

the ball turns and goes a new way.



Machines that push and pull

for teachers and families

[sciencemuseum.org.uk](https://www.sciencemuseum.org.uk), objects and stories

Your turn

★ Watch me find out 4

Can a tap from the side turn the ball?

- 1 Rook rolled the marble straight along the smooth board.
- 2 Bram tapped it once from the left side.
- 3 The marble turned to the right and rolled off the board.

We found out: a push from the side changed the direction of the marble

- 1 Roll a marble along a board. Draw the way it went.
- 2 Tap it once from the side. Draw the new way it went.
- 3 Was your tap a push or a pull?
- 4 Name a game where you turn a ball with your hand.
- 5 What happens to a ball that hits a wall?
- 6 Draw an arrow to show the way your marble went first.
- 7 Say one way you turn a barrow round a corner.

How do you know?

Nib says only a big force can change direction. What do you think?
How do you know?

✦ Nib's notebook

A rook turns in the air with one wing. It tips a little way over and the wind does the rest.

Your turn, and finding out together

1



2



3



Three things that roll. Every one of them can be turned.

8 Write the name of each thing above.

9 Which of the three rolls furthest? Try it and find out.

10 Roll a ball to a friend. Ask her to tap it back.

11 Draw the path your ball took. Use one arrow for each push.

12 Build a wall of books. Roll the ball at it. What happens?

13 Now roll the ball along the wall. Draw the new path.

14 Say which force turned your ball best.

▲ The material you need

a marble, a wooden ball, a smooth board and four books for a wall

Find out together

Build a path with books on the bench, with two corners in it. Roll the marble in at one end. Use taps from the side to turn it round every corner.

! Take care

Keep marbles off the floor. A marble under a shoe can send you over.

UNIT 2 · LOOKING BACK

Words we used

■ WORDS WE USED



push

a force that moves something away from you



pull

a force that moves something towards you



force

a push or a pull



squash

to press something into a flatter shape



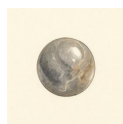
bend

to make something curve without breaking it



slope

a surface that is higher at one end



speed

how fast something is moving



direction

the way something is going

■ Check what you can do

- 1** Push a book across your table, then pull it back. Name each force.
- 2** Squash a ball of clay flat and say what you changed about it.
- 3** Make a ramp steeper and say what happens to a marble on it.
- 4** Roll a ball, then stop it with your hand. Say what your hand did.
- 5** Bat a ball sideways and say how its direction changed.

UNIT 2 · LOOKING BACK

Now I can

- ☐ name a push and a pull when I see one
- ☐ change the shape of something by squashing, bending or stretching it
- ☐ make something go faster with a bigger push
- ☐ make a moving thing change direction
- ☐ say that a force is a push or a pull

How did you do?

	confident	almost there	needs more work
I can name a push and a pull	☐	☐	☐
I can change the shape of a material	☐	☐	☐
I can make something go faster	☐	☐	☐
I can change which way something goes	☐	☐	☐
I can say what a force is	☐	☐	☐



One crate, pushed away and pulled back.

▲ For teachers

Pupils at this age often think a force is only a push. The pulling half of Topic 2.1 needs as much time as the pushing half, and the twine activity is the one that makes it stick, because the pupil feels the pull in her own paws.

The ramp in Topic 2.3 is also Step 2 of the year-long project. Keep the plank and the bricks: the class will need them again, and reusing the same apparatus helps pupils see the two lessons as one enquiry.

Getting materials right

Moss has tipped the odds box out on the bench. Now she is sorting every piece into heaps.



On the bench

Put one hand on your table and one hand on your jumper. What is each one made of? Which one is softer?

○ What Moss noticed

The odds box has wool, twigs, jars, tins and old cloth in it. Moss wants to sort the lot. Where does she start?

Moss needs a rule before she starts. Every good sort begins with one question about the things themselves.

● By the end you will

- sort materials into natural and made
- name the properties of a material
- choose the right material for a job
- plan a fair test on two materials
- change a material by squashing, bending or tearing



Moss sorts the odds box into two heaps.



3.1 Natural and made materials

WAKE UP
YOUR
THINKING

Look at your shoe. Point to a part that came from a plant or an animal. Now point to a part that people made.

▲ Let us find out

Everything around you is made of something. That something is a **material**.

Some materials are **natural**. Nobody made them. Wool comes from a sheep. Wood comes from a tree. Stone and shell are natural too.

Some materials are **made** by people. Glass, tin, paper and cloth are all made in a works or a mill.

A made material often starts as a natural one. Paper starts as wood. Cloth can start as wool.

Moss's own notes at the bench

material	where it came from
wool	from a sheep
wood	from a tree
stone	from the ground
shell	from a sea animal
glass	made by people
tin	made by people

**What things are made of**

for teachers and families

nhm.ac.uk, discover**Your turn**★ **Watch me find out 1****Is this paper natural or made?**

- 1 Moss looked at a twig from the hedge. Nobody had made it.
- 2 She looked at a sheet of paper. It was flat, white and the same all over.
- 3 Nib told her that paper is pressed from wood in a mill.

We found out: the paper is a made material, because people press it out of wood

- 1 Write natural or made next to each one: wool, glass, twig, tin.
- 2 Which animal gives us wool?
- 3 Which natural material does paper start as?
- 4 Name one natural material in this room.
- 5 Name one made material in this room.
- 6 Draw a shell. Write where a shell comes from.
- 7 Bram says stone is made by people. Is he right? Say why.

How do you know?

Rook says wool must be a made material, because his cape is made of it. Is he right? How do you know?

✦ Nib's notebook

A sheep grows a new coat of wool every year. So one sheep can give us wool again and again.

Your turn, and sorting for real

1



2



3



4



Four things from the odds box. Two are natural and two are made.

8 Write the name of the material under each picture above.

9 Write natural or made under each name.

10 Find five things on your table. Sort them into two heaps.

11 Draw your natural heap.

12 Draw your made heap.

13 Which heap has more things in it?

14 Find one thing that is hard to sort. Say why it is hard.

▲ The material you need

a box of odds and ends, two paper trays and a hand lens

Find out together

Tip your box out on the bench. Sort every piece into a natural tray and a made tray. Look closely with the hand lens first. Then swap trays with a friend and check each other.



The bench, with a tray for each heap.

! Take care

Handle glass and tin carefully. Tell an adult about a sharp edge, and do not pick it up yourself.



3.2 Properties of materials

Do you remember?

You have sorted materials into natural and made. Now you will say what each one is like. How would you describe a rubber band?

▲ Let us find out

Every material is like something. What it is like is called a **property**.

A material can be **hard** or **soft**. Stone is hard. Felt is soft.

It can be **bendy** or **stiff**. A rubber band is bendy. A ruler is stiff.

It can be **stretchy**. Pull a rubber band and it gets longer.

It can be **waterproof**, so the rain runs off it. It can be **see-through**, like the glass in a jar.

One material can have many properties at once.

Nib felt each material and wrote it down

material	hard or soft?	one more property
stone	hard	not bendy at all
felt	soft	not waterproof
glass	hard	see-through
rubber	soft	stretchy

Your turn

★ Watch me find out 2

Which cloth keeps the rain out?

- 1 Nib laid felt and oilcloth flat on the bench.
- 2 He dripped ten drops of water on each one.
- 3 The felt went dark and wet right through. The drops sat on top of the oilcloth.

We found out: the oilcloth is waterproof, because the water stayed on top of it

- 1 Name one hard material and one soft material.
- 2 Which is stretchy: a rubber band or a ruler?
- 3 Which is see-through: felt or glass?
- 4 Press your finger into a lump of clay. Draw what you see.
- 5 Find something stiff on your table. Write its name.
- 6 Find something bendy on your table. Write its name.
- 7 Write two properties of a glass jar.

How do you know?

Tick says paper is waterproof, because she keeps her notes dry in a paper bag. What do you think? How do you know?

Your turn, and testing with your hands



Four materials to feel: felt, glass, rubber and paper.

- 8 Write soft or hard under each material above.
- 9 Which of the four is see-through?
- 10 Which of the four tears most easily?
- 11 Bend each one gently. Which one springs back?
- 12 Make a list of every material in your pencil case.
- 13 Write one property beside each material on your list.
- 14 Which material in your case is the softest?

▲ The material you need

small pieces of felt, paper, oilcloth and rubber, a dropper and a dish of water

Find out together

Lay each piece flat on the bench. Drip ten drops of water on each one. Watch where the water goes. Which pieces let the water through?

✦ Potting shed challenge

Find a material that is soft and waterproof at the same time. Draw it and write what it is used for.



3.3 Using the right material

WAKE UP
YOUR
THINKING

Think of a window made of wood. What would be wrong with it?

▲ Let us find out

We choose a material for a job. We choose it because of its properties.

A window is made of glass. Glass is see-through, so the light gets in.

A coat for the rain is made of oilcloth. Oilcloth is waterproof.

A ruler is made of stiff wood. It will not bend when you draw.

The wrong material spoils the job. A paper coat would tear in the rain. A wool window would let in no light at all.

Bram's list of jobs at the shed

the job and the material	why we chose it
window: glass	it is see-through
rain coat: oilcloth	it is waterproof
ruler: wood	it is stiff
cushion: felt	it is soft



What things are made of
for teachers and families
nhm.ac.uk, discover

Your turn

★ Watch me find out 3

What should Fen make the seed tray from?

- 1 Fen made one tray from paper and one tray from tin.
- 2 She filled both trays with wet soil and left them a week.
- 3 The paper tray went soggy and sagged. The tin tray held its shape.

We found out: tin is the right material for a seed tray. It stays stiff even when it is wet

- 1 Why is a window made of glass?
- 2 Why is a rain coat made of oilcloth?
- 3 Name a good material for a cushion. Say why.
- 4 Name a bad material for a bucket. Say why.
- 5 Draw a bridge for a toy car. Write what you would build it from.
- 6 Which material would you pick for a plant pot?
- 7 Look at your chair. Say why it is made of that material.

Look a little closer

A greenhouse is made of glass all over. Write two reasons why the glass suits the job.

How do you know?

Moss says she will mend her boot with paper, because paper is easy to cut. Is that a good plan? How do you know?

Your turn, and choosing for real



Four jobs at the shed. Each one needs the right material.

8 Write the material each thing above is made of.

9 Write one reason for each choice.

10 Choose a material for a bag to carry wet soil.

11 Say why you chose it.

12 Draw a hat for a rainy day. Label the material.

13 Would a wool hat keep you dry? Say what you think.

14 Find one thing at school made of the wrong material.

▲ The material you need

scraps of paper, felt, oilcloth and cloth, string, and a small stone to carry

Find out together

Make a small bag from one scrap. Put the stone inside and carry it round the room. Then hold the bag under a dripping tap. Which scrap makes the best bag?

✦ Nib's notebook

The first umbrellas were made of paper. They were oiled with tree sap, so rain would run off them.



3.4 Testing materials

Do you remember?

You have felt materials with your hands. Now you will prove what they are like.
What makes a test fair?

▲ Let us find out

We can **test** a material to find out what it is like.

A good test is a **fair test**. In a fair test you change one thing only. You keep everything else the same.

Fen tested three cloths for the rain. She dripped ten drops on each one. She used the same dropper and the same ten drops.

Only the cloth changed. So the test was fair.

If Fen used twenty drops on one cloth, the test would not be fair. Nobody could trust the answer.

One thing changed, and one thing only

in Fen's test	kept the same	changed
the dropper	yes	no
ten drops of water	yes	no
the cloth	no	yes
the time she waited	yes	no

Your turn

★ Watch me find out 4

Which paper is the strongest?

- 1 Rook hung a cup on a strip of each paper.
- 2 He used strips of the same width and the same length.
- 3 He dropped in one marble at a time. The thin paper tore after two marbles. The card held nine.

We found out: the card is the strongest paper. It held the most marbles in the same test

- 1 How many things do you change in a fair test?
- 2 Name one thing Rook kept the same.
- 3 Why did he use strips of the same length?
- 4 Rook drops two marbles in one cup and six in another. Is that fair?
- 5 Write down the one thing you would change in your own test.
- 6 Write down two things you would keep the same.
- 7 Draw the cup hanging on a paper strip.

How do you know?

Nib pours a jug of water on the felt and one drop on the oilcloth. He says the oilcloth wins. Is that fair? How do you know?

Your turn, and a fair test of your own

The same dropper, the same ten drops, three cloths.

WHAT I THINK WILL HAPPEN

Draw or write which cloth you think, before you start

WHAT REALLY HAPPENED

Draw or write what really happened

Which cloth will keep the water out best?

- 8** Write down which cloth you think will win.
- 9** Drip ten drops on each cloth. Use the same dropper each time.
- 10** Which cloth stayed driest underneath?
- 11** Was it the cloth you thought? Compare what happened with your guess.
- 12** Say why you used the same number of drops on each cloth.
- 13** Name one thing that would spoil your test.
- 14** Draw your three cloths in order, from driest to wettest.

▲ The material you need

three cloths, a dropper, a dish of water and dry paper to lay underneath

Find out together

Lay each cloth on a dry sheet of paper. Drip ten drops on each one. Lift the cloth and look at the paper. Keep the dropper and the drops the same.

✦ Potting shed challenge

Test the same three cloths again with twenty drops. Does the same cloth win?



Five ways to change a thing with your hands

3.5 Changing materials

WAKE UP
YOUR
THINKING

Squash a ball of paper in your fist. Now open your hand. Is the paper the same as before?

▲ Let us find out

You can **change** a material with your hands.

You can **squash** it, so it gets flatter. You can **bend** it into a curve.
You can **twist** it round.

You can **stretch** it longer. You can **tear** it into pieces.

Some changes go back. A rubber band springs back to its old shape.

Some changes stay. Torn paper stays torn. You cannot untear it.

Goes back:

a bent ruler, a stretched rubber band, a squashed sponge, a twisted cloth

Stays changed:

torn paper, squashed clay, a snapped twig, a bent tin lid

**What happens to what we throw away**

for families

gov.uk, recycling collections

Your turn**★ Watch me find out 5****Does a rubber band go back to its old shape?**

- 1 Tick measured a rubber band on the bench. It was six centimetres long.
- 2 She pulled it hard and held it. It was now eleven centimetres long.
- 3 She let go and measured it again. It was six centimetres long.

We found out: the rubber band goes back to its old shape after stretching

- 1 Name the five ways to change a material.
- 2 Which one makes a material longer?
- 3 Which one makes a material flatter?
- 4 Draw a twisted cloth.
- 5 Tear one corner off a scrap of paper. Try to mend it.
- 6 Write one change that stays.
- 7 Write one change that goes back.

How do you know?

Fen says every change can be undone. What do you think? How do you know?

✦ Nib's notebook

Old glass jars can be melted and made into new jars again and again, and nothing is wasted.

Your turn, and mending at the bench



Moss bends a strip of tin to mend the broken tray.

- 8 Squash, bend, twist, stretch and tear a scrap of cloth.
- 9 Draw the cloth after each change.
- 10 Which change was the hardest to do?
- 11 Which of your changes went back on its own?
- 12 Roll clay into a ball. Now squash it flat. Can you get the ball back?
- 13 Find a bent thing in the shed. Draw it.
- 14 Say one thing your class could mend instead of throwing away.

▲ The material you need

clay, a rubber band, a strip of cloth, scrap paper and a ruler

Find out together

Try to change each material five ways. Squash it, bend it, twist it, stretch it and tear it. Write down which ways worked for each material.

! Take care

Do not pull hard on a rubber band near your face. Let it go slowly.

UNIT 3 · LOOKING BACK

Words we used

■ WORDS WE USED



natural

it comes from a plant or an animal



made

people make it in a works or a mill



material

the stuff a thing is made of



soft

it presses in when you push it, like felt



stretchy

it gets longer when you pull it



waterproof

water stays on top and does not soak in



see-through

you can look right through it



tear

to pull something into pieces

■ Check what you can do

- 1** Pick up two things from your table and say which is natural.
- 2** Hold a jar up to the window and name two of its properties.
- 3** Choose a material for a shed roof and say why it suits the job.
- 4** Plan a fair test on two cloths, and say what you keep the same.
- 5** Change a scrap of paper two ways, and say which change stays.

UNIT 3 · LOOKING BACK**Now I can**

- ☐ sort materials into natural ones and made ones
- ☐ name the properties of a material, and use the right word
- ☐ choose a material for a job, and say why it suits it
- ☐ plan a fair test that changes one thing only
- ☐ change a material by squashing, bending, twisting or tearing

How did you do?

	confident	almost there	needs more work
I can sort materials into natural and made	☐	☐	☐
I can name the properties of a material	☐	☐	☐
I can choose the right material for a job	☐	☐	☐
I can say what makes a test fair	☐	☐	☐
I can change a material and say what happened	☐	☐	☐



Moss's finished sort, with a reason under every heap.

▲ For teachers

Pupils sort readily by what a thing IS, and less readily by where it came from. The paper card in Topic 3.1 is the useful hard case: it is made, yet it begins as wood. Let pupils argue about it, then show them a wood shaving and a sheet of paper side by side.

Topic 3.4 is the fair test, and the water drop test carries it. The point is not which cloth wins, it is that the dropper and the ten drops never changed. Ask pupils what they would say to a friend who used a whole jug on one cloth.

Humans and animals grow

Nib is watering the seedlings on the windowsill, and one of them is taller than his ruler.



On the bench

Hold your hand flat against the wall. Was your hand ever smaller than that? How do you know?

○ What Nib noticed

I put twelve bean seeds in the tray on Monday. By Friday two of them were taller than the others. Why are they not all the same?

Nib is right to wonder. Every seed came out of the same packet. Something has made two of them grow faster.

● By the end you will

- match a young animal to its parent
- measure a plant and record how much it grows
- say why a lamb has wool and a duckling has a bill
- name five things that keep your body healthy
- name the teeth that cut and the teeth that grind



One living thing, drawn at four different ages.



Four young animals, and four grown-up parents

4.1 Comparing animals

WAKE UP
YOUR
THINKING

Think of a baby animal you have seen. Was it bigger or smaller than its mother?

▲ Let us find out

Many animals have young. A young animal grows into an adult like its parents.

A hen has chicks. A cat has kittens. A duck has ducklings.

A young animal is smaller than its parent. It is often softer too.

A chick is covered in fluffy **down**. A hen has proper feathers. A kitten has big eyes and small ears.

Put the two animals side by side. Then you can see what is the same, and what has changed.

Nib's list of pairs from the farm next door

adult	its young	one thing that is different
hen	chick	the chick has soft down
cat	kitten	the kitten has bigger eyes
duck	duckling	the duckling is much smaller
sheep	lamb	the lamb has thin legs

Your turn★ **Watch me find out 1****Which of these two animals is the young one?**

- 1 Nib looked at a hen and a chick together in the run.
- 2 The chick stood beside the hen. Its head reached her knee.
- 3 The chick was covered in soft yellow down. The hen had firm feathers.

We found out: the chick is the young one. It is smaller, and its feathers are still soft down

- 1 Name the young animal that a hen looks after.
- 2 Name the young animal that a duck looks after.
- 3 Which is taller: a sheep or a lamb?
- 4 Write one way a kitten is different from a cat.
- 5 Draw an adult animal. Draw its young beside it.
- 6 Label which one of your two drawings is older.
- 7 Name a young animal you have really seen. Say where you saw it.

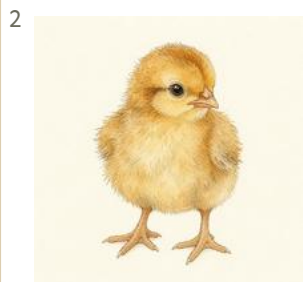
How do you know?

Nib says a small cat must be a kitten, because kittens are small. Is he right? How do you know?

★ Nib's notebook

A newly hatched duckling can swim on its first day. A chick of the same age must stay dry on land.

Your turn, and looking closely



Two adults and two young animals. Which two belong together?

8 Write the name under each animal above.

9 Draw a line to join each young animal to its parent.

10 Write adult or young under each of the four.

11 Which two of the four have a flat bill?

12 Which two of the four are covered in down?

13 Sort these into adult or young: lamb, cat, duckling, hen.

14 Nib says every young animal has feathers. Say why he is wrong.

▲ The material you need

photographs of farm animals with their young, a hand lens and a pencil

Find out together

Look at a photograph of an adult animal and its young. List three things that are the same. Then list two things that are different.

✦ Potting shed challenge

A tadpole does not look like a frog at all. Draw a tadpole, and draw the frog it will become.



Nib on his stool, ruler against the tallest bean

4.2 Growing

Do you remember?

You know that a young animal is smaller than its parent. So what must happen to the young animal next?

▲ Let us find out

All living things **grow**. You were once a baby. Now you are much taller.

Growing takes time. It happens slowly, so you cannot watch it happen.

We can **measure** growth. Nib measures his seedlings every Monday. He writes each height in his notebook. The numbers get bigger week by week.

A measurement beats a guess. It gives you proof that something has grown.

Four Mondays in Nib's notebook

Monday	height of Nib's bean	how much it grew
week 1	2 cm	the first shoot showed
week 2	6 cm	it grew 4 cm
week 3	11 cm	it grew 5 cm
week 4	15 cm	it grew 4 cm



Grow more than beans

for teachers

rhs.org.uk, school gardening

Your turn

★ Watch me find out 2

Is the bean on the windowsill really growing?

- 1 On Monday Nib stood his ruler beside the bean. It reached 6 cm.
- 2 He did not touch the bean for a whole week.
- 3 The next Monday he used the same ruler in the same place. It reached 11 cm.

We found out: the bean is really growing. It gained 5 cm in one week on the same ruler

- 1 How tall was Nib's bean in week 1?
- 2 How tall was it in week 4?
- 3 In which week did the bean grow most?
- 4 Why did Nib use the same ruler every Monday?
- 5 Measure your own hand span in centimetres. Write it down.
- 6 Measure a friend's hand span. Whose is longer?
- 7 Name one part of you that has grown this year.

How do you know?

Nib says his bean grew overnight, because it looks taller today.
What do you think? How do you know?

Your turn, and a fair test**WHAT I THINK WILL HAPPEN**

Draw or write which one you think, before you start

WHAT REALLY HAPPENED

Draw or write what really happened

Which seedling will be taller after one week?

- 8 Write down which seedling you think will win.
- 9 Measure both seedlings today. Write both heights down.
- 10 Water both seedlings the same amount every day.
- 11 Measure both again after seven days.
- 12 How many centimetres did each one grow?
- 13 Compare what really happened with what you thought first.
- 14 Say one reason your two seedlings grew different amounts.

▲ **The material you need**

two seedlings in pots, a ruler, a small watering can and a notebook

Find out together

Stand your ruler in the soil beside each seedling. Measure on the same day each week. Keep every height in your notebook, so you can see the growing.

✦ **Potting shed challenge**

Nib forgot to water one pot for a week. What do you think the ruler showed then?



4.3 Inheriting characteristics

WAKE UP
YOUR
THINKING

Look at somebody in your family. Name one thing about them that looks like you.

▲ Let us find out

A young animal looks like its parents. We say it has **inherited** the way it looks.

A lamb has woolly hair, because a sheep has wool. A duckling has a flat bill, because a duck has one.

Young animals in one family are not all the same. One kitten may be grey and one may be ginger.

You look a little like the people in your family. You are not exactly the same as any of them.

Nib's notes from the farm gate

young animal	what it inherited
lamb	wool, like its mother
duckling	a flat bill, like its mother
kitten	whiskers, like its mother
chick	two legs and a beak, like its mother

Your turn

★ Watch me find out 3

Whose lamb is the one with the black face?

- 1 Two ewes stood in the field. One had a white face and one had a black face.
- 2 Nib watched the black-faced lamb all morning.
- 3 It stayed close to the ewe with the black face.

We found out: the black-faced lamb belongs to the black-faced ewe. Young animals inherit the way their parents look

- 1 Why does a lamb have wool?
- 2 Why does a duckling have a flat bill?
- 3 Name one thing a kitten inherits from its mother.
- 4 Draw a cat. Now draw its kitten with the same colour coat.
- 5 Are two kittens in one family always the same colour?
- 6 Write one thing you have inherited from your family.
- 7 Write one thing about you that nobody else in your family has.

How do you know?

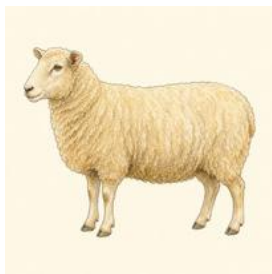
Nib says a duckling could grow up into a hen, because both are birds. Is he right? How do you know?

✦ Nib's notebook

Some lambs are born black and turn creamy white as they grow. A shepherd cannot always tell by colour alone.

Your turn, and sorting for real

1



2



3



4



Two families. Join each young animal to the parent it resembles.

8 Write the name under each animal above.

9 Join each young animal to its parent with a line.

10 Which pair has wool?

11 Which pair has whiskers?

12 Draw the ears of the adult cat and the ears of the kitten.

13 Write one way the two pairs of ears are the same.

14 Ask five friends about eye colour. Make a tally of the answers.

▲ The material you need

a mirror, paper and coloured pencils

Find out together

Look in the mirror. Draw your own eyes and your own hair. Then ask at home which grown-up has eyes the same colour.

★ Potting shed challenge

Nib has grey fur and so does his mother. Does that prove every wood mouse is grey? Say what you think.



Fruit, water, a blanket, a rope and a bar of soap

4.4 Keeping healthy

Do you remember?

You know that your body grows. What do you think your body needs, so that it can grow well?

▲ Let us find out

Your body needs care every day. Five simple things keep you healthy.

You need **food** of many kinds. Fruit and vegetables help you most.

You need **water** to drink. Your body loses water all day long.

You need **sleep**. Your body mends itself while you rest.

You need **exercise**. Running and skipping make your heart strong.

You need to **wash**. Soap lifts the germs off your hands.

Every day you need:

different foods, water to drink,
enough sleep, time to run about,
clean hands

Too much of these is not good:

sweet drinks, sweets and cake,
sitting still all day, late nights

**Food, sleep and moving about**

for families

nhs.uk/healthier-families

Your turn**★ Watch me find out 4****Does washing your hands really take the dirt off?**

- 1 Nib rubbed soil on both paws after digging.
- 2 He rinsed one paw in plain water. Some soil stayed in the fur.
- 3 He washed the other paw with soap and warm water. That paw came out clean.

We found out: soap and water clean better than water alone. The soap lifted the soil off the fur

- 1 Name the five things your body needs every day.
- 2 Which drink is best for your body?
- 3 Why do you need to sleep?
- 4 Name one game that makes your heart beat faster.
- 5 Draw a plate. Draw two healthy foods on it.
- 6 When should you always wash your hands?
- 7 Write one thing you will do today to keep healthy.

How do you know?

Nib says a big bowl of sweets will make him strong, because sweets give him energy. What do you think? How do you know?

Your turn, and keeping a record

My healthy day

- 8 Write what you ate for breakfast today.
- 9 Count how many cups of water you drink in one day.
- 10 Write the time you went to sleep last night.
- 11 Count how many minutes you ran about at playtime.
- 12 Draw the exercise you like best.
- 13 Keep your record for three days. Which day was healthiest?
- 14 Say one thing you would change next week.

▲ **The material you need**

soap, warm water, a paper towel and a hand lens

Find out together

Wash your hands with soap for as long as it takes to count to twenty. Look at your nails with the hand lens before and after. What has changed?

! **Take care**

Only ever eat or drink what your teacher or your family gives you. Never taste anything you do not know.



A brush, a mirror and one apple with a bite out of it

4.5 Teeth

WAKE UP
YOUR
THINKING

Run your tongue along your own teeth. Are they all the same shape?

▲ Let us find out

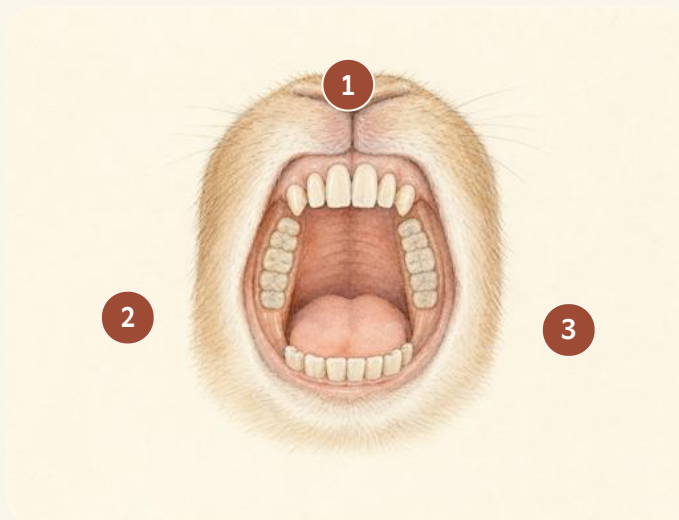
Your **teeth** are hard and white. They do more than one job.

The front teeth are flat and sharp. They cut into your food.

The back teeth are wide and bumpy. They grind the food small.

You grew baby teeth first. Those teeth fall out, and adult teeth grow in their place.

Look after every tooth. Brush twice a day, and eat few sweet things.



- | | |
|---------------|----------------|
| 1 front teeth | 4 2 side tooth |
| 3 back teeth | 4 gum |

This set of teeth has 4 parts labelled. Find each one in a mirror.



Looking after your teeth

for families

nhs.uk, healthy teeth and gums

Your turn

★ Watch me find out 5

Which teeth do the cutting?

- 1 Nib took one bite of apple and stopped.
- 2 He looked at the mark left in the apple. It was a neat straight line.
- 3 The straight line matched the flat sharp teeth at the front of his mouth.

We found out: the front teeth do the cutting. Their flat edge matched the mark in the apple

- 1 Which teeth are flat and sharp?
- 2 Which teeth grind your food small?
- 3 What do your back teeth feel like?
- 4 Draw a front tooth. Draw a back tooth beside it.
- 5 How many times a day should you brush your teeth?
- 6 Name one food that is kind to your teeth.
- 7 Say what happens to a baby tooth when it falls out.

How do you know?

Nib says he can chew a hard nut with his front teeth. Is he right?
How do you know?

★ Nib's notebook

A wood mouse has front teeth that keep growing all its life, because gnawing wears them down.

Your turn, and counting for real

What Nib's class said about brushing

What we found	Tally	How many
brushes twice a day		18
brushes once a day		7
not sure		3
altogether		28

8 How many in Nib's class brush twice a day?

9 How many brush only once?

10 How many children answered altogether?

11 Ask your own class the same question. Make a tally.

12 Count your own teeth in the mirror. Write the number down.

13 Draw the gap where a tooth has come out.

14 Write two rules for looking after teeth.

▲ The material you need

a mirror, a toothbrush of your own and a slice of apple

Find out together

Bite the apple slice once with your front teeth. Then chew a piece with your back teeth. Write down which teeth cut, and which teeth grind.

! Take care

Use only your own toothbrush, and never share it. Wash your hands before you touch food.

UNIT 4 · LOOKING BACK

Words we used

■ WORDS WE USED



young

a new animal that will grow into an adult



adult

a grown-up animal



grow

to get bigger over time



measure

to find out how big something is



inherit

to get the way you look from your parents



healthy

well, and looked after



germs

tiny living things that can make you ill



grind

to crush food small with your back teeth

■ Check what you can do

- 1** Point to a young animal in a picture and name its parent.
- 2** Measure a plant with a ruler and say how much it grew.
- 3** Say why a lamb has wool and a duckling has a flat bill.
- 4** Sort ten pictures of food into everyday and once in a while.
- 5** Show which of your teeth cut, and which of them grind.

UNIT 4 · LOOKING BACK**Now I can**

- ☐ match a young animal to its parent, and say what is different
- ☐ measure a plant each week and record how much it has grown
- ☐ say why young animals look like their parents
- ☐ name five things that keep my body healthy
- ☐ name the teeth that cut and the teeth that grind

How did you do?

	confident	almost there	needs more work
I can match a young animal to its parent	☐	☐	☐
I can measure how much a plant grows	☐	☐	☐
I can say what a young animal inherits	☐	☐	☐
I can name five things that keep me healthy	☐	☐	☐
I can name the different kinds of teeth	☐	☐	☐



Nib measuring the tallest bean on the windowsill.

▲ For teachers

Pupils accept that things grow, and forget that growing must be measured. Nib's notebook in Topic 4.2 is the argument that carries it: same ruler, same day of the week, numbers that rise. Repeat the weekly measure with a real seedling if a pupil is unconvinced.

Topic 4.3 invites family talk, so tread gently. Keep to what can be seen: wool, bills, whiskers, eye colour. Two kittens of different colours in one family are the useful example, because they show that inheriting is not copying.

Light

Fen holds up the lit lantern in the shed doorway at dusk, and the whole floor turns gold.



On the bench

Look around this room. How many things can you see that make their own light? Which one is the brightest?

○ What Fen noticed

I shut the shed door to keep the rain out, and I could not see my own pots. Where did they go?

Fen is right to wonder. The pots did not go anywhere. Without light we cannot see them at all.

● By the end you will

- name things that make their own light
- say why the Moon is not a light source
- say what darkness is, and why we need light to see
- say what makes a shadow on the ground
- measure how a shadow changes through one day



Light sources in the shed, and one thing that only borrows its light.



Four lights in one small shed

5.1 Light sources

WAKE UP
YOUR
THINKING

Point to something in this room that makes its own light.
Now point to something that only looks bright.

▲ Let us find out

Some things make their own light. We call them **light sources**.

The **Sun** is a light source. It lights the whole sky in the day.

A candle is a light source. So is a lamp, a torch and a fire.

Fen's lantern is a light source too. It makes its own light inside the glass.

The **Moon** is not a light source. It has no light of its own. The Moon looks bright because it **reflects** the Sun's light.

A mirror is not a source either. It only sends light back to us.

Fen's own notes

thing	does it make its own light?
the Sun	yes, a light source
a candle	yes, a light source
a torch	yes, a light source
the Moon	no, it reflects



Why the days get longer

for teachers

[metoffice.gov.uk](https://www.metoffice.gov.uk), learn about weather

Your turn

★ Watch me find out 1

Is the mirror a light source?

- 1 Fen shut the mirror inside a dark box. The mirror went black.
- 2 He shut the torch inside the same box. The torch still shone.
- 3 A mirror has no light of its own. It can only send light back.

We found out: the torch is a light source. The mirror is not, because it went black in the dark box

- 1 Name three things that make their own light.
- 2 Which one is a light source: a mirror or a candle?
- 3 Draw Fen's lantern. Colour the light around it.
- 4 Write yes or no: is the Moon a light source?
- 5 Say why the Moon looks bright at night.
- 6 Name the biggest light source we have.
- 7 Name one light source in your own home.

How do you know?

Nib says a mirror must be a light source, because it looks so bright.
Is he right? How do you know?

Your turn, and finding out together

Four objects from the shed. Only three of them make light.

- 8 Write the name under each object above.
- 9 Tick every object that makes its own light.
- 10 Which object above is not a light source?
- 11 Count the light sources in your classroom.
- 12 Draw the brightest one you found.
- 13 Which gives more light: a torch or a candle? Say what you think.
- 14 Name one light people used long ago, before lamps.

▲ The material you need

a torch, a small mirror, a shoebox with a lid and a sheet of grey paper

Find out together

Stand the mirror in the shoebox and shut the lid. Is the mirror bright now? Lift the lid a little and shine the torch in. Watch what the mirror does with the light.

! Take care

Never shine a torch into anybody's eyes. Let an adult light every candle.



The shed with the door shut, and one lantern lit

5.2 Darkness

Do you remember?

You know which things make their own light. What happens in a room when you take every light away?

▲ Let us find out

Darkness is what we have when there is no light.

We need light to see. In the dark we cannot see colour or shape.

Shut the shed door and the shed grows dark. Open it again and the light comes flooding back in.

Your eyes do help you a little. In a dark room they slowly get used to it.

After a while you begin to see grey shapes. Even then you cannot see colour properly.

Bring in one small light and every colour comes back.

Fen's notes from the potting shed

where Fen stood	light or dark?	what he could see
shed, door open	light	every pot on the shelf
shed, door shut	dark	only grey shapes
shed, lantern lit	light	every pot again
deep under the bench	dark	almost nothing at all

Your turn★ **Watch me find out 2****Can you see colour in the dark?**

- 1 Fen put a red pot and a blue pot on the bench.
- 2 He shut the shed door. Both pots looked grey.
- 3 He lit the lantern. The red pot looked red again.

We found out: we need light to see colour. In the dark both pots looked the same grey

- 1 What is darkness?
- 2 Why can you not read in a dark room?
- 3 Name one place that is dark all day long.
- 4 Draw the shed with its door shut. Draw it again with the door open.
- 5 Write what you can see in each of your drawings.
- 6 What could Fen do to see inside the dark shed?
- 7 Do your eyes change in a dark room? Say what you think.

How do you know?

Moss says her eyes make a little light of their own in the dark. Is she right? How do you know?

★ Nib's notebook

An owl can hunt in light so poor that we would see nothing. Its eyes are huge for the size of its head.

Your turn, and looking into the dark

WHAT I THINK WILL HAPPEN

Draw or write what you think, before you look in

WHAT REALLY HAPPENED

Draw or write what really happened

What will you be able to see inside the dark box after two minutes?

- 8** Write down what you think you will see in the dark box.
- 9** Look in through the small hole for two minutes.
- 10** Write down what you can see now.
- 11** Compare what you really saw with what you thought.
- 12** Now shine the torch in. Write what you can see.
- 13** Which came back first: the shapes or the colours?
- 14** Say why a lantern helps Fen in the dark shed.

▲ The material you need

a large box with a lid, a dark cloth, three coloured pots and a torch

Find out together

Put the three coloured pots inside the box. Cover the box with the cloth. Look in through one small hole. Wait two minutes and look again. Which pot can you name?

! Take care

Never put a box or a bag over your head. Keep the box on the bench.



5.3 The Sun appears to move

WAKE UP
YOUR
THINKING

Look at the floor near the window. Can you see a shadow?
What is blocking the light?

▲ Let us find out

A **shadow** forms when something blocks the light.

Stand in the sunshine and your body blocks the light. A dark shape falls on the ground beside you.

Shadows do not stay the same all day. They change in **length** and in **direction**.

In the early morning shadows are long. Near the middle of the day they are short.

In the evening they grow long again, and they point the other way.

The Sun **appears to move** across the sky. That is why our shadows swing slowly round.

The Earth is the thing that really moves. It travels round the Sun, and it spins once each day.

So watch the shadows on the ground. They tell you the story safely.

Fen's shadow notes from one sunny day

the time	how long was the shadow?	which way did it point?
9 o'clock	long	towards the shed
12 o'clock	short	towards the wall
3 o'clock	long again	towards the gate



Staying safe in the sun

for teachers and families

nhs.uk, sunscreen and sun safety

Your turn

★ Watch me find out 3

Why did the shadow move?

- 1 Fen stood a straight stick in a pot of sand on the path.
- 2 At 9 o'clock he chalked round the shadow. It was long.
- 3 At 12 o'clock he chalked it again. It was short, and it pointed a different way.

We found out: the shadow grew shorter and swung round. The Sun appears to move across the sky

- 1 What makes a shadow?
- 2 Draw a post in the sunshine. Draw its shadow too.
- 3 When are shadows longest: in the morning or at midday?
- 4 Which way did Fen's shadow point at 9 o'clock?
- 5 Does the shadow of a wall move? Say what you think.
- 6 Why are there no sharp shadows on a cloudy day?
- 7 Name one thing that blocks the light in your classroom.

How do you know?

Rook says his shadow will stay in the same place all day. Is he right? How do you know?

★ Nib's notebook

The longest shadows of the whole day fall just after sunrise and just before sunset.

Your turn, and a whole day of shadows



Fen's shadow clock: one stick, one pot of sand and three chalk marks.

- 8** Stand a straight stick in a pot of sand on a sunny path.
- 9** Chalk round its shadow in the morning. Write the time beside it.
- 10** Chalk round the shadow again at midday. Write the time.
- 11** Chalk round it once more in the afternoon.
- 12** Measure each shadow with a ruler. Write the three lengths.
- 13** Which of your three shadows was the longest?
- 14** Draw your three marks. Show which way each one pointed.

▲ The material you need

a straight stick, a pot of sand, chalk, a ruler and a sunny patch of ground

Find out together

Set your stick in the pot on a sunny path. Mark the shadow three times in one day. Keep the pot in exactly the same place all day long.

! Take care

Watch the shadows on the ground, and never look straight at the Sun. Wear a hat on a hot day, and drink some water.

UNIT 5 · LOOKING BACK

Words we used

■ WORDS WE USED



light source

something that makes its own light



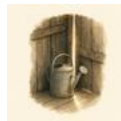
lantern

a lamp that holds its light inside glass



torch

a small light you can carry about



darkness

what we have when there is no light



reflect

to send light back again



shadow

a dark shape where something blocks the light



shadow clock

a stick that shows the time by its shadow



length

how long something is

■ Check what you can do

- 1** Point to two light sources in this room and name them both.
- 2** Say why the Moon looks bright, even though it makes no light.
- 3** Tell a friend what darkness is, in your own words.
- 4** Block a torch beam with your hand and name what you have made.
- 5** Say how a shadow changes between the morning and midday.

UNIT 5 · LOOKING BACK

Now I can

- ☐ name things that make their own light
- ☐ say why the Moon is not a light source
- ☐ say what darkness is, and why we need light to see
- ☐ say what makes a shadow on the ground
- ☐ measure a shadow, and say how it changed

How did you do?

	confident	almost there	needs more work
I can name a light source	☐	☐	☐
I can say why the Moon is not a light source	☐	☐	☐
I can say what darkness is	☐	☐	☐
I can say what makes a shadow	☐	☐	☐
I can measure how a shadow changes	☐	☐	☐



Fen at the shed door at dusk, with the lantern held high.

▲ For teachers

Pupils happily list lamps and candles, then add the Moon and a mirror. The dark box test in Topic 5.1 settles it without argument: the torch still shines, the mirror goes black. Repeat it with any shiny object a pupil offers.

Topic 5.3 is the unit's safety point as well as its science. Keep every pupil looking down at the chalk marks, never up at the sky. Say that the Sun appears to move, and that the Earth is the thing really travelling round it.

Electricity

Tick pressed the little button by the shed door. The bell rang at the far end of the bench.



On the bench

Look around this room. How many things can you see that use electricity? How do you know they use it?

○ What Tick noticed

I pressed the button by the shed door. The bell rang right at the other end of the bench. Nothing moved in between.

Tick is right to wonder. No string pulled the bell and no paw touched it. Something ran along the thin wire instead.

● By the end you will

- name things that use electricity
- say what electricity does for us
- say the rule about mains electricity
- keep water well away from electricity
- make a circuit that lights a bulb



Tick's bell, its wire and the button by the shed door.



6.1 Where do we use electricity?

WAKE UP
YOUR
THINKING

Point to something in this room that needs electricity.
Now point to something that does not need it at all.

▲ Let us find out

Electricity makes things work. It runs along wires to the thing that needs it.

Electricity does five different jobs for us.

It makes **light**, so a lamp can shine. It makes sound, so a radio can play. It makes **heat**, so a kettle can boil. It makes cold, so a **fridge** can keep the milk fresh. It makes things move, so a fan can turn.

Some things take their electricity from the wall. Others carry a **battery** inside them.

A battery holds a little electricity. The wall holds a great deal of it.

Tick's chart: what does the electricity do?

what it is	light	sound	heat	it moves
lamp	☐	☐	☐	☐
radio	☐	☐	☐	☐
kettle	☐	☐	☐	☐
fan	☐	☐	☐	☐
torch	☐	☐	☐	☐
bell	☐	☐	☐	☐

Your turn★ **Watch me find out 1****What does the electricity do in each thing?**

- 1 Tick switched on her torch. It made a bright light.
- 2 She pressed the button by the door. The little bell made a sound.
- 3 She switched on the small fan. It moved the air across the bench.

We found out: electricity can make light, sound and movement, and it made all three on one bench

- 1 Name three things in this room that use electricity.
- 2 What does the electricity do in a lamp?
- 3 What does the electricity do in a radio?
- 4 Which one gets hot: a fan or a kettle?
- 5 Draw one thing at home that uses electricity.
- 6 Write what that thing does for you.
- 7 Name one thing that works with a battery.

How do you know?

Tick says the shed door needs electricity to open. Is she right? How do you know?

★ Nib's notebook

The very first electric doorbell was made over a hundred years ago. It worked in just the same way as Tick's bell does.

Your turn, and counting for real



Four things that use electricity: a lamp, a radio, a kettle and a fan.

8 Write the name under each thing above.

9 Which of the four makes a sound?

10 Which of the four makes heat?

11 Which of the four moves the air?

12 Count the things in this room that use electricity.

13 Draw the one you would miss the most.

14 Say why you would miss it.

▲ The material you need

a clipboard, a sheet of paper and a pencil

Find out together

Walk round your classroom with your partner. Write down every thing you find that uses electricity. Look, but do not touch. Then count your list.

Look a little closer

A lamp needs electricity, and so does a lighthouse. Name one big thing outside that needs electricity all night long.



Two things only an adult may touch

6.2 Keep safe with electricity

Do you remember?

You know that some things take their electricity from the wall. Do you know who is allowed to work there?

▲ Let us find out

The electricity in the wall is called **mains** electricity. It is very strong indeed.

Mains electricity can kill you. It is never a thing to play with.

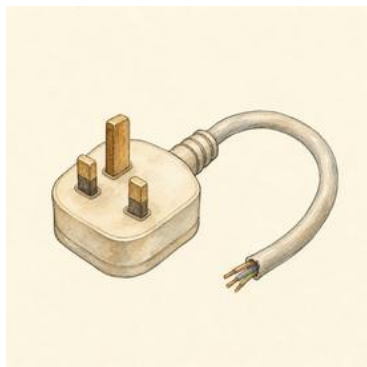
Only an adult may put a **plug** into a **socket**. Only an adult may pull one out again. That is the rule, and it does not change.

Water carries electricity too. So never bring water near a socket, and never touch a switch with wet hands.

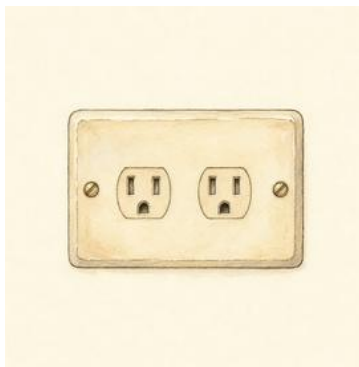
A **battery** is not like the wall. A battery is small and weak, so you may use one yourself.

That is why every test in this book uses a battery.

1



2



3



Two of these are for adults only. The third one is safe for you.

**Take care**

Mains electricity can kill. Only an adult may plug in or unplug a mains appliance. Never poke anything into a socket. Never bring water near a socket, a wire or a switch.

Your turn**Watch me find out 2****Which lamp may Tick switch on by herself?**

- 1 Tick found two lamps on the bench. One had a long wire that went to the wall.
- 2 The other one was small, with a battery inside it.
- 3 The wall lamp takes mains electricity, and mains is for adults only.

We found out: Tick may switch on the battery lamp by herself. An adult must help with the other one

- 1 Who is the only one allowed to work at the wall?
- 2 Why must you keep water away from electricity?
- 3 Draw a battery. Write its name under it.
- 4 Why is a battery safe for you to use?
- 5 Name one thing at home that only an adult may switch on.
- 6 Say one place in your home where you must take care.
- 7 Who would you tell if you saw a broken wire?

How do you know?

Fen says his paws are dry, so he may switch on the big lamp at the wall. What do you think? How do you know?

**Electrical safety at home**

for families

[electricalsafetyfirst.org.uk](https://www.electricalsafetyfirst.org.uk), safety advice

Your turn, and finding out together

Mains electricity is very strong.	TRUE / NOT TRUE
A battery is safe for you to use.	TRUE / NOT TRUE
Water is safe near electricity.	TRUE / NOT TRUE
You may switch on a wall lamp on your own.	TRUE / NOT TRUE

- 8 Tick TRUE or NOT TRUE for each sentence above.
- 9 Write the rule about water in your own words.
- 10 Draw a warning card. Show wet hands and a wire on it.
- 11 Draw the shed torch. Write why it is safe to hold.
- 12 Moss says water cannot carry electricity. Is she right? Say why.
- 13 Name two things that work with a battery.
- 14 Say what you would do if a wire looked broken.

▲ The material you need

a torch, one battery for it and a dark corner of the room

Find out together

Work with a partner. Put the battery into the torch and switch it on. Now take the battery out again and try the switch. What happens without the battery?

★ Potting shed challenge

A torch has a battery, so it works in a power cut. Name two more things in your home that would still work.



One battery, two wires and a bulb that lights

6.3 Making circuits

WAKE UP
YOUR
THINKING

Hold hands in a ring with three friends. Now let one pair of hands go. Is the ring still whole?

▲ Let us find out

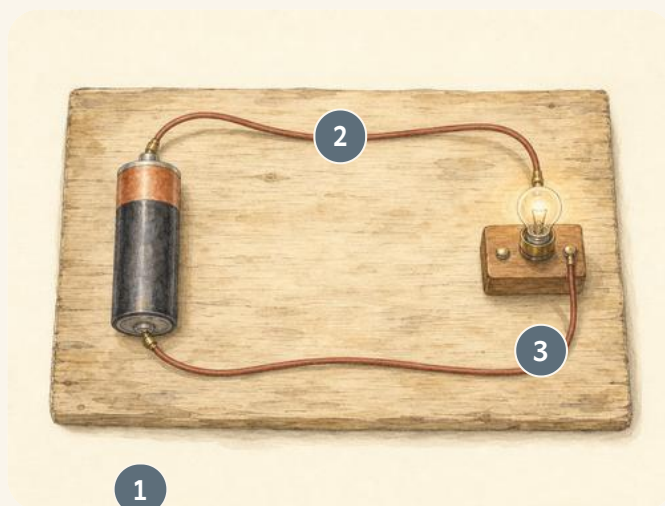
A **circuit** is a loop. The electricity runs all the way round it.

You need three things to light a bulb. You need a **battery**, two **wires** and a **bulb**.

Join the battery to the bulb with the two wires. Now the loop is closed, so the bulb lights up.

Take one wire off again. There is a gap in the loop now. The bulb goes out at once.

A **switch** makes that gap on purpose. Press the switch and the loop closes again.



- | | |
|-----------|----------|
| 1 battery | 4 2 bulb |
| 3 wire | 4 switch |

This circuit has 4 parts labelled. Find each one in the box your teacher gives you.

Your turn

★ Watch me find out 3

Why did the bulb go out?

- 1 Tick made a loop with a battery, two wires and a bulb. The bulb lit up.
- 2 She pulled one wire off the battery. The bulb went out straight away.
- 3 She pressed the same wire back on. The bulb lit up again.

We found out: the bulb lights only when the loop is complete. A gap stops the electricity

- 1 Name the three things you need to light a bulb.
- 2 What is a circuit? Say it in your own words.
- 3 Draw a loop with a battery and a bulb in it.
- 4 What happens to the bulb when there is a gap?
- 5 Which part of a circuit holds the electricity?
- 6 Which part carries the electricity round the loop?
- 7 What does a switch do to the loop?

How do you know?

Rook says a bulb will light with one wire and no battery. Is he right? How do you know?



Build another circuit
for teachers
education.theiet.org, primary

Your turn, and a fair test

WHAT I THINK WILL HAPPEN

Draw or write what you think, before you start

WHAT REALLY HAPPENED

Draw or write what really happened

Will your bulb shine brighter with one battery or with two?

- 8** Write down what you think will happen when you open the switch.
- 9** Make your loop with the battery, two wires and the bulb.
- 10** Draw your circuit. Label the battery and the bulb.
- 11** Take one wire off. Write what the bulb does.
- 12** Compare what really happened with what you thought.
- 13** Put the switch into your loop. Press it, then let it go.
- 14** Say what the switch does, in your own words.

▲ The material you need

one battery in a holder, two wires with clips, one small bulb in a holder and one push switch

Find out together

Work with a partner and use the battery only. Build the loop and light the bulb. Then swap your bulb for a second one and look again. Which bulb shines the brighter?

! Take care

Use the battery only for this work. Never take a bulb or a wire to the wall. Mains electricity can kill, and only an adult may work there.

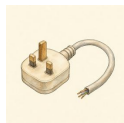
UNIT 6 · LOOKING BACK

Words we used

■ WORDS WE USED

**electricity**

the power that makes many things work

**mains**

the strong electricity in the walls of a building

**battery**

a safe store of electricity you can hold

**bulb**

it makes light when electricity flows through it

**wire**

electricity flows along it

**switch**

it opens or closes a circuit

**circuit**

a complete loop for electricity to flow round

**socket**

where a plug goes, and only an adult may use it

■ Check what you can do

- 1 Point to three things in this room that use electricity.
- 2 Say what electricity is doing in each one: light, sound or heat.
- 3 Say the rule about plugs and sockets, in your own words.
- 4 Build a circuit that lights a bulb, using a battery only.
- 5 Break your circuit in one place and say why the light goes out.

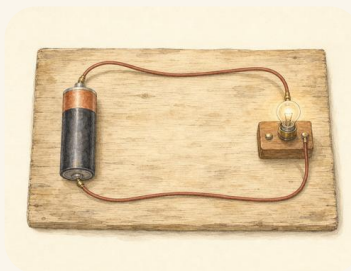
UNIT 6 · LOOKING BACK

Now I can

- ☐ name things that use electricity, and say what it does in them
- ☐ say why mains electricity is only ever for an adult to touch
- ☐ build a circuit with a battery, wires and a bulb
- ☐ make a switch, and say what it does to the loop
- ☐ say why a broken loop gives no light

How did you do?

	confident	almost there	needs more work
I can name things that use electricity	☐	☐	☐
I know the rule about plugs and sockets	☐	☐	☐
I can build a circuit that works	☐	☐	☐
I can make and use a switch	☐	☐	☐
I can say why a broken circuit gives no light	☐	☐	☐



One loop, one battery, one bulb, and it lights.

▲ For teachers

The safety rule in Topic 6.2 is the one sentence in this book that must be word perfect: only an adult may plug in or unplug a mains appliance, and mains electricity can kill. Say it, have pupils say it back, and return to it every time the batteries come out.

All pupil circuit work in this unit uses a battery, and no activity anywhere in the book goes near a socket. If a pupil asks why the bulb is dimmer than a room lamp, that is the answer: a battery is a small, safe store, and the wall is not.

Step 1: our habitat map



A habitat map, with real pressed leaves from each place.

STEP 1 OF 6**► Our own garden guide**

You have found four kinds of place in your grounds. Now make a map of them, so the rest of the class can find them too.

- 1 Draw the outline of your grounds on the big sheet. Keep it simple.
- 2 Mark each habitat you found: the shady place, the dry place, the wet place and the open place.
- 3 Glue one pressed leaf or one small stone from each place onto its own part of the map.
- 4 Draw one animal you really saw in each place.
- 5 Put your map in the class folder. It is page one of your guide.

▲ The material you need

one large sheet of paper, glue, coloured pencils, and one pressed leaf or small stone from each habitat

! Take care

Wash your hands after handling soil, leaves or stones.

Step 2: our rolling ramp



A plank, two bricks, and three things that roll.

STEP 2 OF 6

► Our own garden guide

A bigger push makes a thing go faster. Build a ramp and prove it.

- 1 Rest one end of the plank on a single brick. That is your low ramp.
- 2 Let each thing roll from the very top. Do not push it.
- 3 Put a wooden label in the ground where each one stopped.
- 4 Now rest the plank on two bricks, so the ramp is steeper.
- 5 Roll each thing again. Mark where it stops with a second label.
- 6 Draw your ramp and both sets of marks for your guide.

▲ The material you need

one plank, two bricks, a stone marble, a wooden ball, a cork and six plain wooden labels

✦ **Potting shed challenge**

Which one travelled furthest from the steeper ramp? Say why you think that happened.

Step 3: our materials chart



Six materials tied on with twine, ready to test.

STEP 3 OF 6

► Our own garden guide

Make a chart of real materials, so anybody can feel them for themselves.

- 1 Cut six small squares, one of each material.
- 2 Tie each square onto the plank with a short piece of twine.
- 3 Feel each one. Is it hard or soft? Bendy or stiff?
- 4 Drip one drop of water onto each square in turn.
- 5 Which squares let the water through? Draw a ring beside those.
- 6 Your chart is page three of the guide.

▲ The material you need

small squares of felt, glass, tin, wood, rubber and paper, a plank, garden twine and a jar of water



Take care

An adult must cut the tin and handle the glass. Never press on a cut edge.

Step 4: our growing record



Five pots, five plants, and a notched stick for each.

STEP 4 OF 6

► Our own garden guide

Living things grow. This step takes five weeks, so start it now.

- 1 Fill five pots with the same compost, level to the top.
- 2 Sow one bean in each pot, at the same depth.
- 3 Stand all five on the same windowsill. Water them the same.
- 4 Each week, stand a stick beside one plant and cut a notch at its height.
- 5 After five weeks, line the sticks up. The notches show how it grew.
- 6 Draw your five sticks for the guide.

▲ The material you need

five terracotta pots, compost, five broad bean seeds, a watering can, five plain sticks and a safe way to cut a notch

▲ For teachers

This is the book's only multi-week investigation, and the notched sticks matter: they make growth a physical object a pupil can hold and compare, rather than a number in a table. Keep the sticks in the class folder alongside the map from Step 1.

Sow spares. Not every bean germinates, and a pupil whose pot stays bare needs one to measure.

Step 5: our shadow clock



A pin, a stone, and three chalk marks through one day.

STEP 5 OF 6**► Our own garden guide**

A shadow moves through the day. Catch it three times and you have a clock.

- 1 Stand the pin upright in the middle of the flat stone.
- 2 Put the stone on the paving in a sunny spot. Do not move it again.
- 3 In the morning, chalk a line where the pin's shadow falls.
- 4 At midday, chalk a second line. In the afternoon, chalk a third.
- 5 Measure each shadow with a ruler. Which was shortest?
- 6 Draw your three marks for the guide.

▲ The material you need

a flat stone, an upright wooden pin, chalk, a ruler and a sunny piece of paving

! Take care

Look only at the shadow on the ground. Never look straight at the Sun, not even for a moment, and never through anything.

Step 6: our circuit board



A battery, a bulb, a bell and a switch on one plank.

STEP 6 OF 6

► Our own garden guide

Build a circuit that really works, and a switch to turn it on.

- 1 Hold the battery onto the plank with two loops of twine.
- 2 Join one wire from the battery to the bulb holder.
- 3 Push the two brass pins into the plank, a thumb's width apart.
- 4 Join the bulb to one pin, and the other pin back to the battery.
- 5 Rest the paperclip across both pins. The bulb lights.
- 6 Lift the paperclip. The loop is broken, so the light goes out.

▲ The material you need

a plank, a battery, twine, two lengths of wire, a bulb in a holder, two brass drawing pins and one metal paperclip



Take care

Use the battery only. Never put a wire, a pin or anything else into a wall socket. Only an adult may plug in or unplug anything.

Our own garden guide



Six steps, sewn into one book.

All year you have made one page at a time. Look in the class folder. You already have six pieces of work:

- | | |
|--------|---|
| Step 1 | your habitat map, with real pressed leaves |
| Step 2 | your rolling ramp, and where each thing stopped |
| Step 3 | your materials chart, with six squares to feel |
| Step 4 | your growing record, and five notched sticks |
| Step 5 | your shadow clock, with three chalk marks |
| Step 6 | your circuit board, with a switch that works |

Nobody else in the world has made this book. It is a record of one year in one garden, by you.

THE WHOLE YEAR

► Our own garden guide

Now put the six pieces together, in order, and sew them into one book.

- 1 Lay your six pieces of work in order, from Step 1 to Step 6.
- 2 Put a plain cream sheet at the front. That is your cover.
- 3 Draw your own garden on the cover. Do not write on it yet.
- 4 Hold the whole pile together and fold it once, down the middle.
- 5 An adult makes three holes along the fold with a bradawl.
- 6 Sew through the holes with garden twine. Tie the ends at the back.
- 7 Now write the title on the cover, and your own name under it.

▲ The material you need

your six pieces of work, two plain cream sheets, garden twine, a large needle, and an adult with a bradawl

Find out together

Swap books with a partner. Read theirs right through. Find one thing in it that you did not find in your own garden. Tell them what it was.

▲ For teachers

The binding is deliberately simple so that every pupil finishes. If time is short, three staples along the fold work as well as sewing, and the pupil still holds a book at the end of it.

Keep one finished guide back for the following year. Showing a real example made by real pupils is worth more than any instruction, and it shows the incoming class where the six steps were leading.

What a scientist does

The five steps, one more time

1

Ask a question

2

Say what you
think will happen

3

Talk about what
to do

4

Look carefully
and record

5

Talk about what
you found

You used these five steps in every unit of this book. You used them on the woodlice and on the ramp. You used them on the six squares of material and on the beans. You used them on the shadow and on the bulb.

That is what makes you a scientist. Not knowing the answer already, but knowing how to go and find it.



The bench, ready for the next question.

Being fair, and saying what you found

A test is **fair** when you change one thing only. In Unit 3 you dripped the **same** water onto **every** square. In Unit 4 you gave all five beans the **same** drink.

If you had watered one bean more, you could not say what made it grow.

What we did

We poured the same water onto each of the six squares.

What we saw

When you have finished, say what you found. Say it plainly, and say how you know. If it surprised you, say that too.

A scientist who says "I do not know yet" is telling the truth. The truth is where the next question comes from.

▲ For teachers

The fair-test idea is the single most transferable thing in this book and the hardest at this age. Pupils will happily change two variables and then argue about which one mattered. When that happens, ask them to run it again changing only one, and let the disagreement do the teaching.

Return to this page whenever a pupil says a result is unfair. The vocabulary here is deliberately the vocabulary of Units 3 and 4, so they can point at their own work as the example.

WORD LIST

Every word we learnt

Every science word in this book, in the order of the alphabet. The name beside each one tells you where you met it.

A

adult a grown-up animalUnit 4

B

battery a safe store of electricity
you can holdUnit 6

bend to make something curve
without breaking itUnit 2

Bram the badger. He sows the
seeds.Getting set up

bulb it makes light when
electricity flows through itUnit 6

C

circuit a complete loop for
electricity to flow roundUnit 6

D

WORD LIST · CONTINUED

darkness what we have when there is no lightUnit 5

direction the way something is goingUnit 2

E

electricity the power that makes many things workUnit 6

dish for water, soil or seedsGetting set up

environment everything around a living thingUnit 1

F

Fen the toad. He carries the light.Getting set up

force a push or a pullUnit 2

G

germs tiny living things that can make you illUnit 4

grind to crush food small with your back teethUnit 4

grow to get bigger over timeUnit 4

H

habitat the place where a plant or animal livesUnit 1

hand lens makes small things look biggerGetting set up

hard rock you cannot scratch it easilyUnit 1

healthy well, and looked afterUnit 4

WORD LIST

Every word we learnt, continued

I

inherit to get the way you look
from your parentsUnit 4

J

jar for holding what you
findGetting set up

L

lantern a lamp that holds its light
inside glassUnit 5

length how long something isUnit
5

light source something that
makes its own lightUnit 5

litter rubbish dropped where it
does not belongUnit 1

M

WORD LIST · CONTINUED

made people make it in a works or a millUnit 3

magnet for finding out what it pullsGetting set up

mains the strong electricity in the walls of a buildingUnit 6

material the stuff a thing is made ofUnit 3

measure to find out how big something isUnit 4

Moss the hedgehog. She knows every material.Getting set up

N

natural it comes from a plant or an animalUnit 3

Nib the wood mouse. He waters and measures.Getting set up

P

pull a force that moves something towards youUnit 2

push a force that moves something away from youUnit 2

R

reflect to send light back againUnit 5

Rook the rook. He pushes and pulls.Getting set up

ruler for measuring how longGetting set up

WORD LIST

Every word we learnt, continued

S

see-through you can look right through itUnit 3

seed tray for growing things inGetting set up

shade a place the sun does not reachUnit 1

shadow a dark shape where something blocks the lightUnit 5

shadow clock a stick that shows the time by its shadowUnit 5

shallow not deepUnit 1

slope a surface that is higher at one endUnit 2

socket where a plug goes, and only an adult may use itUnit 6

soft it presses in when you push it, like feltUnit 3

soft rock it crumbles or scratches easilyUnit 1

speed how fast something is movingUnit 2

squash to press something into a flatter shapeUnit 2

store to keep something for laterUnit 1

stretchy it gets longer when you pull itUnit 3

switch it opens or closes a circuitUnit 6

T

tear to pull something into piecesUnit 3

Tick the wren. She makes the bell ring.Getting set up

torch for making light in the darkGetting set up

WORD LIST

Every word we learnt, continued

W

watering can for giving plants the same drink
Getting set up

waterproof water stays on top and does not soak in
Unit 3

wire electricity flows along it
Unit 6

Y

young a new animal that will grow into an adult
Unit 4

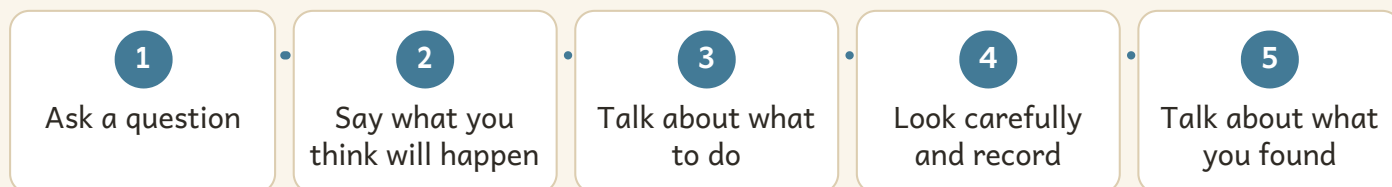
About this book

Who this is for. Science & Lab Year 2 is written for pupils of six and seven, following the Prime School primary science programme. Twenty-five topics run across six units, and every topic carries a modelled enquiry, at least fourteen things to do, and a "How do you know?" prompt.

Safety. Practical work assumes direct adult supervision. Carry out your own risk assessment first and follow your school policy. Two rules are worth repeating aloud at every session rather than reading once: never look straight at the Sun (Unit 5), and only an adult may plug in or unplug a mains appliance, because mains electricity can kill (Unit 6). All pupil circuit work in this book uses a battery, and no activity anywhere goes near a socket.

Misconceptions this book tackles head on. That the Moon makes its own light (Unit 5, where it is named as a reflector). That the Sun moves round the Earth (Unit 5 teaches the Sun's APPARENT movement, through shadows on the ground). That a force is only ever a push (Unit 2 gives pulling equal time, with twine the pupil feels in her own paws). That a fair test can change two things at once (Unit 3, where the disagreement is left to do the teaching). In each case the wrong idea is named, put in a character's mouth, and then tested, because a misconception that is never spoken aloud is never given up.

The five enquiry steps, as pupils meet them



▲ For teachers

The five steps above are generated from one list shared with the front matter and the closing skills pages, so all three always match.

Assessment is built in rather than bolted on. The check-your-progress questions at the end of each unit are pitched to be answered aloud, and the 'Now I can' page beside them is for the pupil to complete, not the teacher. A pupil who ticks a statement they cannot yet demonstrate has told you something useful.

Where to go next

The garden in this book is one garden. These three codes take you outside it. They lead to places that keep records no printed page can hold. Those records change every week of the year.



Ponds, and what lives in them

freshwaterhabitats.org.uk



Name the tree your leaf came from

woodlandtrust.org.uk, british trees



Join a real butterfly count

bigbutterflycount.butterfly-conservation.org

▲ For teachers

These three are chosen because each one is a LIVING dataset rather than a reference page: a pond habitat survey, a tree identification key that changes with the season, and a national butterfly count whose results arrive each August. A printed book cannot carry any of them, which is the test every code in this book has to pass.

The butterfly count runs in high summer. If your term ends before it opens, use the previous year's results map, which stays online, and compare it with what pupils recorded in their own habitat map from the Unit 1 project.

PRIME SCHOOL PRESS

Science

Year 2 · Prime School Press · Student Manual

How do you know?

Come up the garden to the old potting shed, where a badger in a work jacket and a wood mouse in a green coat want to know how everything works.

Real science for children of six and seven. Find out why a woodlouse chooses one stone and not another, how tall a bean grows in five weeks, and how to make a bulb light up.

INSIDE THIS BOOK

- Twenty-five topics across six units
- Over 350 things to do, say and draw
- A modelled enquiry in every single topic
- Six projects, and a garden guide to finish
- QR codes that open real science beyond the page
- A word list of every science word you meet

Prime School Press · Science

Ages 6–7 · Lower Primary

primeschool.pt